

India's Semiconductor Industry

Problem Statement

India imports >80% of its semiconductor requirements, exposing it to supply chain disruptions and geopolitical risks.

TEAM

Aditi

Analysis &
Presentation

Arpit

Analysis

Dhairya

Research

Shreya

Presentation
Development

What Is the Industry?

Semiconductors are the fundamental building blocks of modern digital technologies — powering smartphones, computers, automobiles, AI, robotics, and AR/VR.

Global Value Chain (Fragmented):

- Chip Design — USA, UK
- Fabrication (Foundries) — Taiwan, South Korea
- Equipment Manufacturing — Netherlands, Japan
- Assembly, Testing & Packaging (ATMP) — Southeast Asia

The Problem

Strategic Risk

Chips are critical for defense, telecom, AI, and quantum computing.

Economic Leakage

Heavy imports increase India's electronics trade deficit and limit domestic high-value manufacturing.

Supply Chain Vulnerability

COVID-19 and geopolitical tensions exposed how semiconductor shortages halt auto & electronics production.

>80%

of India's
semiconductors imported

\$300B+

global semiconductor
market size

₹91,000Cr

Tata–PSMC fab
investment in Dholera

Economic & Strategic Significance

- Semiconductors are no longer just technological enablers — they are strategic assets influencing national security, economic resilience, and global supply chain dynamics.
- **India's Govt Focus:** India Semiconductor Mission (ISM) & ISM 2.0 aim to develop domestic fabrication, packaging, and design capabilities through multi-billion dollar incentive schemes.
- **Cutting-edge domains** like quantum computing, AI, robotics, and AR/VR will continue to depend on rapid semiconductor innovation — sectors where India aspires to lead.
- **Geopolitical constraints:** Concentration in Taiwan, South Korea, USA & China; trade restrictions and export controls create systemic risk for import-dependent economies.

Framework	Primary Purpose	Key Questions It Answers	Why Appropriate?
PESTEL	Macro-level Eagle Eye View	Political trade barriers? Economic incentives? Social talent pool? Tech readiness? Environmental standards? Legal IP protection?	Analyses the macro-level factors affecting the semiconductor ecosystem in India
SWOT	Strategic Recommendations	What are India's strengths in chip design? Weaknesses in fabrication? Opportunities from China+1? Threats from incumbents?	Enables clear and effective strategic recommendations for India's path
Value Chain	Industry Dynamics	Where does India currently fit in the global value chain? Where are the highest-value nodes? Where are the gaps?	Critical to understanding future trends and India's integration opportunities
Porter's 5 Forces	Competitive Landscape	Bargaining power of chip suppliers? Rivalry among foundries? Threat from China's expansion? Buyer concentration?	Decodes competition dynamics and India's positioning challenges

Primary Research (Planned)

Stakeholder Interviews — Minimum 4

Status: Awaiting contacts from Prof. Sarthak — 1 interview per team member

Secondary Data Sources

Government & Policy

- India Semiconductor Mission (PIB)
- ISM 2.0 Official Documentation

Industry Reports

- IESA Workforce Strategy Report 2025
- IBEF — India Semiconductor Hub Emergence
- Dec 2024: 'Semiconductor Problem of India'

Trade & Financial Data

- Company Financials: Tata, Micron, Vedanta
- RBI & Commerce Ministry Statistics

Academic Papers

- Peer-reviewed semiconductor economics
- Supply chain resilience studies

India Almost Became a Semiconductor Hub in the 1980s

Semiconductor Complex Limited (SCL) was established in Chandigarh — before Taiwan or Korea scaled. A major fire in 1989 and policy delays prevented India from capitalising on this early-mover advantage.

Private Sector Momentum — Major Conglomerates Entering

Tata Group, Vedanta, and HCL have announced significant investments in fabrication, design, and packaging, driven by government incentives and global demand.

Tata–PSMC: India's First Large Fab in Dholera, Gujarat

₹91,000 crore investment in partnership with Taiwan's Powerchip Semiconductor Manufacturing Corporation — marking a historic milestone in domestic fabrication.

Micron Technology: Assembly & Testing Plant in Gujarat

Investment exceeding ₹22,000 crore for ATMP (Assembly, Testing, Packaging) facilities — aligning with India's more competitive position in this segment of the value chain.

Timeline & Milestones

Phase	Milestone	Owner
Week 1–2	Proposal presentation + feedback incorporation	All
Week 3–4	Primary research — stakeholder interviews (4–5 interviews)	All (1 each)
Week 5–6	Secondary data collection & framework analysis	Arpit, Dhairya
Week 7–8	PESTEL, SWOT, Value Chain & Porter's analysis	Aditi, Arpit
Week 9–10	Slide development + narrative building	Aditi, Shreya
Week 11	Final review, rehearsal & delivery	All

Hypotheses to Test:

- 1. Incentives alone insufficient for full-fab ecosystem
- 2. Geopolitics creating supply-chain opportunity for India
- 3. India more competitive in design & packaging than fabrication

Division of Responsibilities

Aditi Analysis + Presentation Development + 1 Interview
Arpit Analysis + 1 Interview
Dhairya Research + 1 Interview
Shreya Presentation Development + 1 Interview

Risk Assessment

 Limited stakeholder access → Leverage Prof. Sarthak's network; target 5+ contacts
 Research complexity → Divide by framework; weekly sync-ups to align